



MONASH
University

FIT3163 | MDS17 | Project Concept and Design Analysis Report

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This assignment is written and curated by members of MDS17_Heart Rate Measurement, to complete the unit and hurdle requirements. Content is tailored according to the specific project details and information, and several research studies were also added to improve the report.

Table of contents

Introduction.....	3
Project goals and expected deliverables.....	3
User Interface Representation.....	4
Welcome, Permissions, and Home screens.....	4
Recording, Processing, Result screens.....	6
History screens.....	7
Conclusion.....	8
Data Analysis Report.....	9
Methodology.....	9
Data Collection Procedure.....	9
Algorithm Training and Data Analysis.....	11
Data Pre-processing.....	12
Data Analysis.....	12
Results and Findings.....	13
Tools and Software specifications.....	14
Justification of Choices.....	16
Tools and Hardware Specifications.....	18
Computer.....	18
Camera.....	20
Heart Rate Measurement Tools.....	20
References.....	21

Introduction

The advent of mobile technology has ushered in a new era of healthcare innovations, enabling sophisticated medical monitoring in everyday settings. Among these advancements, the ability to measure vital signs using non-contact methods stands out for its potential to revolutionise patient care and athletic training. This project focuses on one such vital sign—heart rate—traditionally measured through direct contact methods such as electrocardiography (ECG) which, while accurate, can be cumbersome and impractical outside clinical environments. The burgeoning field of photoplethysmography (PPG) offers a promising alternative, harnessing optical techniques to detect blood volume changes in the microvascular bed of tissue. Our research leverages this technology through the medium most ubiquitous today: the mobile phone camera.

Project goals and expected deliverables

The significance of heart rate measurement extends beyond the medical field into areas such as fitness monitoring and psychological assessment. Conventional methods, while reliable, often require specialised equipment and physical contact, which can be restrictive. The mobile phone, equipped with a camera capable of capturing high-resolution video, presents an attractive tool for achieving non-contact heart rate measurements. It utilises the subtle colour changes in facial skin, corresponding to the blood flow, to ascertain the heart rate. To streamline the complexity of our project, we propose a simplified approach: by instructing users to keep their faces still for 10 seconds, we aim to minimise the impact of motion artifacts and variable lighting conditions, focusing on optimising the data quality under controlled settings. This method, however, is not without challenges—the inherent noise in video data can significantly skew accuracy.

Our project aims to refine mobile-based heart rate measurement using camera capture, focusing on optimising Principal Component Analysis (PCA), Independent Component Analysis (ICA), and Canonical Component Analysis (CCA) to handle mobile constraints. We seek to establish the minimal video duration needed targeting 10 seconds for accurate readings while minimising motion and lighting artifacts, enhancing the practicality of non-invasive health monitoring on mobile devices.

User Interface Representation

Creating an intuitive and user-friendly mobile app interface for a health-related product like a heart rate monitor using video analysis would involve proper design and strategic planning. The ultimate goal is to make the technology accessible and straightforward, ensuring that users of all technical backgrounds can effectively interact with the app without confusion. The following text will delve into the reasoning behind each UI screen's design, justifying design decisions based on user experience principles, and explore how the screens work together to provide a pleasant user experience. This will be a conceptual showcase of what the final product could look like as it is a key component of the design of our project and would lay the groundwork for the rest of our project which will ultimately govern the development process.

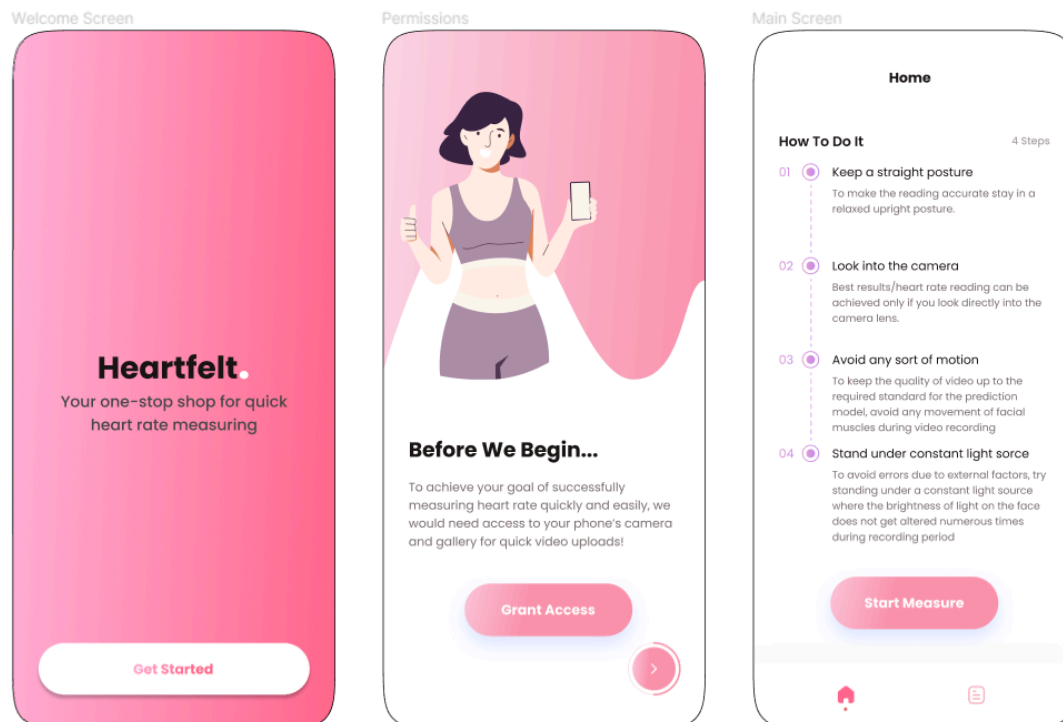


Diagram 1.1: User Interface Flow for Heart Rate Measurement App from Welcome Screen to Home Screen.

Welcome, Permissions, and Home screens

Starting with the "Welcome Screen", as seen on the left-most side of Diagram 1.1, is the first point of interaction between the app and the users. The design of this screen is extremely critical since it is the thing that sets the tone for the entire user experience. A clean and inviting interface with minimal text introduces the app briefly, highlighting its core functionality and benefits without overwhelming the user in the form of a subtext. The choice of a warm,

welcoming colour palette that includes hues of red and pink can be soothing, which is inclined to health-related applications. An easily distinguishable "Get Started" button invites the user to engage immediately with the app. At the same time, an optional "How It Works" link could offer those interested in the technology or methodology a deeper understanding, thereby catering to varying levels of curiosity and technical comfort. However, the button is not depicted in the representation below yet as it is something we would finalise adding during the development process.

Moving on to the Permissions Request Screen, shown at the centre of Diagram 1.1, the necessity for camera access is explained clearly as transparency in why permissions to the user's camera and gallery are needed builds trust, which is rather important in apps where people's facial identity is directly involved and privacy is a major concern. Therefore users are more likely to grant permissions if they fully understand their purpose and benefit, ensuring the functionality of capturing video for heart rate analysis can proceed unhindered. The design remains straightforward, with clear calls to action such as the "Grant Access" and "Proceed" buttons in the form of the right-ward facing arrow, maintaining user engagement and trust.

Once the user proceeds from granting access, they are greeted by the main/home screen, shown at the rightmost of Diagram 1.1, which is meticulously designed as the operational hub where users begin their interaction with the core functionality of the app. A large central "Start Measure" button reduces user effort and decision-making time, which is essential in a consumer-facing product where ease of use drives user satisfaction and retention. Moreover, to ensure that users know how to take heart-rate measures effectively by reducing chances of errors caused due to factors such as motion, posture, and lighting, a succinct 4-step instructional process is effectively outlined right above the call to action i.e. the "Start Measure" button. Putting the instructions as the first thing a user sees on the main screen sets them up on the right track towards taking accurate heart rate readings with minimal chances of errors. The background colour here is chosen to be neutral to minimise distractions while the user focuses on starting the measurement process.

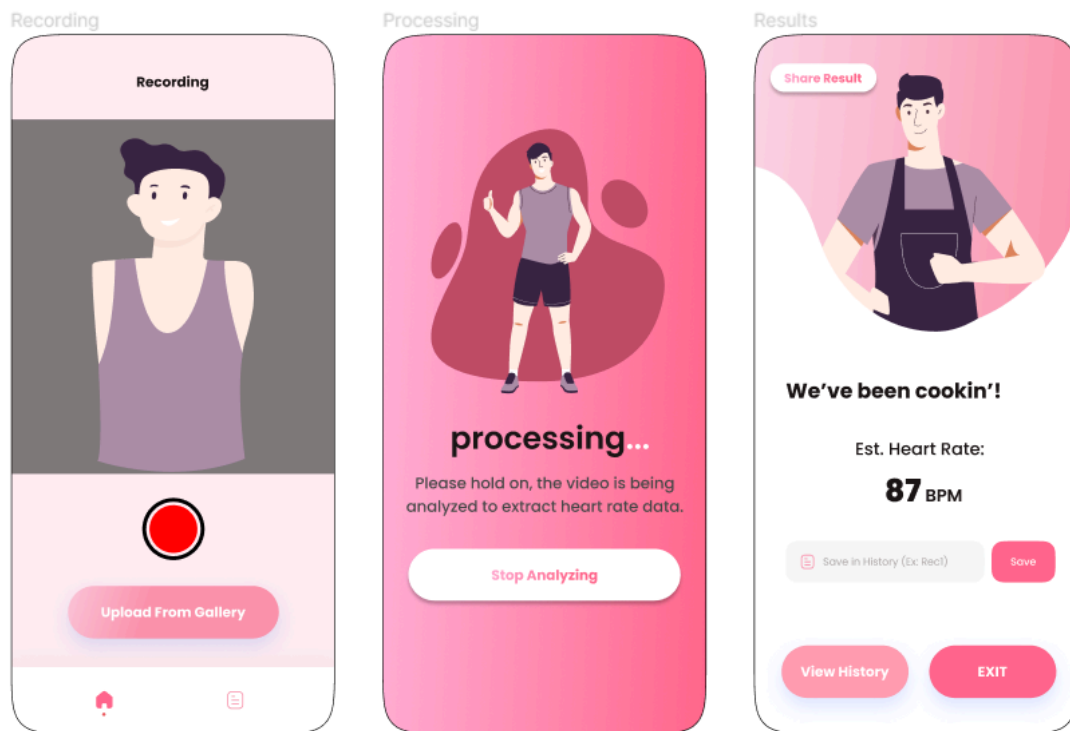


Diagram 1.2: Recording, Processing, and Results Screens of the Heart Rate Measurement App.

Recording, Processing and Result screens

Upon initiating the heart rate measurement, the user would smoothly transition to the Recording screen shown in Diagram 1.2, which features a prominent red-coloured video recording button, and a 10-second countdown timer once the user presses the record button. On top of this, the screen would also feature a button labelled "Upload From Gallery" giving the user a convenient option to measure the heart rate of existing facial videos they had recorded in the past. This design reinforces user autonomy by allowing them to stop the process at any time by clicking the home button on the taskbar at the bottom of the screen, enhancing the app's user-centric approach. The countdown timer also manages expectations and helps users prepare for the brief interaction, minimising anxiety about the duration of data capture.

Once the recording or upload from the gallery is successfully completed, the user is shown a Processing Screen as shown in the centre of Diagram 1.2, which effectively signals to the user that their data is being analysed for estimated heart rate, employing a progress bar/loading animation in the form of tree dots (...) which is crucial for maintaining user engagement during wait times as the model might take some time to display predicted heart rate output. As a result, this screen is highly crucial in managing user expectations, which is a critical factor in user experience, especially in health-related apps where anxiety about

results could be a factor. Moreover, if a user accidentally uploads a wrong video file or does not feel satisfied with their video recorded within the app, they would still be given the option in the form of a "Stop Analysing" button to stop the process in the middle of the analysis by the model, ensuring that convenience and time saving is a top priority.

Following analysis, the Results Screen located in the rightmost of Diagram 1.2 presents the heart rate data in a bold, easy-to-read format, allowing users to quickly grasp the main information. The options to save the reading by giving it a name, retake the measurement (in the form of the "Exit" button), view history, or share results are presented as secondary actions, which supports clear decision-making pathways in the app's workflow. This screen uses colour effectively, with calming colours like blue or green for the background, and contrasting colours for buttons that require attention or action, guiding the user's eye naturally to options without overwhelming them with choices.

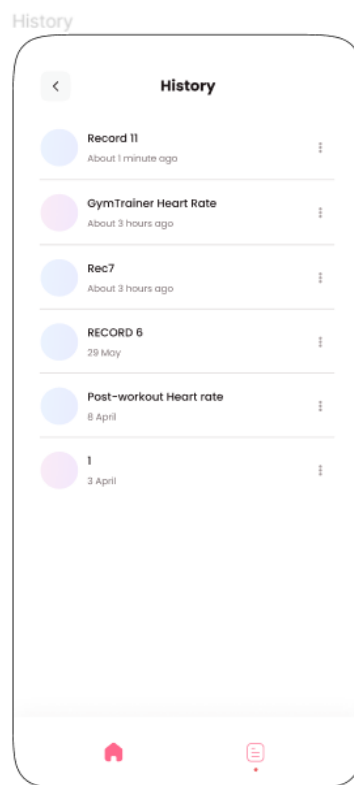


Diagram 1.3: History Screen Interface of the Heart Rate Measurement App.

History screen

Lastly, the History Screen shown in Diagram 1.3, which can be accessed by clicking the "View History" button at the end of each reading or by clicking the icon at the bottom right corner of the screen, offers a retrospective look at past measurements, designed with

simplicity to encourage regular monitoring and engagement with the app. This functionality not only adds value to the user's experience by tracking health over time but also leverages minimal design to keep the user focused on historical data, which is presented in an easy-to-scan format displaying the name of each recording as well as the dates they were taken at.

Conclusion

To conclude the UI representation of our project, intuitive design is not just a matter of aesthetic appeal but a fundamental component that influences the effectiveness of the product. The design decisions made in this app, from colour choices to the flow of screens, are driven by the need to create a seamless, intuitive user journey that minimises cognitive load and maximises user satisfaction and trust. These elements are not just complementary; they are deeply interconnected, each contributing to a holistic user experience that is engaging, and informative. This strategic design approach ensures that the final product of our project not only meets the functional needs of measuring heart rate but also supports a broader goal of promoting health monitoring as accessible and routine.

Data Analysis Report

Methodology

The methodology of this project involves a structured approach to data collection and algorithm training, tailored to refine and validate the heart rate measurement algorithm implemented on mobile devices. This process is critical for ensuring the algorithm's effectiveness across varying physiological conditions and maintaining consistency in less controlled environments. The data collection is conducted in two primary stages—resting and post-exercise—to capture a broad range of heart rates. This methodology section details the steps involved, from initial setup to the final analysis, employing specific software and tools to enhance accuracy and reliability.

Data Collection Procedure

Participant Recruitment and Preparation

1. Selection Criteria:

- **Demographics:** A minimum of 10 participants are selected to represent a diverse demographic spectrum, including different ages, genders, and fitness levels. This diversity helps ensure that the algorithm performs well across a broad user base.
- **Health Screening:** Participants are interviewed for a basic health screening to identify any conditions that might influence heart rate variability, such as cardiovascular diseases or respiratory issues.

2. Preparation Guidelines:

- **Dietary Restrictions:** Participants are instructed to avoid any strenuous activity at least 3 hours before the experiment to standardise initial conditions.

Environment Setup

1. Facility Preparation:

- **Lighting Conditions:** The data collection area is set up with consistent, neutral lighting to avoid variations in video quality that could affect facial recognition accuracy.

- **Camera Setup:** The mobile phone camera is secured on a stable mount at a fixed distance from where participants will be positioned, ensuring consistency in the video captures.

2. Initial Data Capture (Resting State):

- **Positioning:** Each participant undergoes a facial recognition session using the integrated mobile application. They are asked to remain as still as possible for at least 10 seconds.
- **Recording:** The mobile application initiates a 10-second video capture. Participants are asked to remain as motionless as possible during this period to minimise facial movements. This session captures the baseline resting heart rate.

Exercise Protocol

1. Activity Description:

- **Exercise Task:** After the initial resting data capture, participants are asked to perform a moderate-intensity aerobic activity, such as jogging or brisk walking, for 15 minutes. This duration is sufficient to elevate the heart rate significantly above the resting level.
- **Safety Measures:** The exercise is conducted in a safe, enclosed area adjacent to the data collection site, and participants are monitored for any signs of distress during the activity.

2. Post-exercise Data Capture:

- **Immediate Capture:** Immediately following the exercise, participants return to the camera setup. They are given a brief moment (approximately 5-10 seconds) to compose themselves before the second data capture begins.
- **Recording Procedure:** As with the initial capture, the participant is recorded for 10 seconds in a stationary position, facing the camera. This capture is critical as it records the elevated heart rate post-exercise.

Algorithm Training and Data Analysis

1. Data Handling

- **Software Utilisation:** The data from both capture sessions are uploaded to a secure server where preprocessing is conducted to align and normalise the inputs for training. The software tools involved include OpenCV for image processing and TensorFlow for implementing and training the deep learning models.
- **Facial Recognition Enhancement:** Techniques such as frame averaging and filtering are applied to enhance the signal quality of the heart rate measurements derived from facial recognition data.

2. Algorithm Training

- **Machine Learning Framework:** The preprocessed data are used to train a convolutional neural network (CNN), designed to recognize and predict heart rate variations from facial image data accurately. Training involves adjusting weights through backpropagation, based on the heart rate data captured at rest and post-exercise conditions.
- **Validation and Testing:** The trained algorithm is validated using a cross-validation approach to ensure its robustness and reliability. The performance metrics, including accuracy, sensitivity, and specificity, are calculated to assess the model's effectiveness.

3. Interpretation and Refinement

- **Feedback Loop:** Based on the testing results, the algorithm undergoes iterative refinements to address any biases or inaccuracies observed. This stage may involve tweaking the neural network architecture or retraining the model with additional data.
- **Final Analysis:** Comprehensive statistical analysis is conducted to compare the algorithm's output with traditional heart rate measurement methods using a watch or wristband to validate its accuracy and efficiency.

Data Pre-processing

Before initiating algorithm training, it's imperative to prepare the captured data through a meticulous pre-processing phase. This process ensures that the input data is optimised for accuracy and consistency, crucial for effective machine learning.

- **Data Scope:** The collected dataset comprises facial video records from 10 participants. Each participant contributes two video records: one in a resting state and another post-exercise, totaling 20 video records.
- **Video Specifications:** Each video is a 10-second clip captured at a resolution suitable for facial recognition processing, ensuring sufficient detail for detecting subtle changes in facial coloration due to heart rate variations.
- **Data Types:**
 - **Video Data:** Composed of sequential image frames captured from the mobile device's camera. Each frame is processed to extract colour intensity values across the RGB spectrum.
 - **Timestamps:** Each video is accompanied by a timestamp indicating the capture time, crucial for aligning pre- and post-exercise data.
 - **Participant Metadata:** Includes anonymized IDs, age, gender, and timestamps of video captures. This metadata helps in stratifying the data during analysis to determine if demographic factors influence heart rate measurements.

Each step in the pre-processing phase is meticulously documented, and changes to the data are tracked with version control systems. This documentation ensures reproducibility and transparency in how the data is prepared before training the machine learning models

Data Analysis

Data Consistencies, Missing Values, Limitations, and Biases

As this project aims to estimate dynamic heart rates using various methods such as PCA, ICA, and CCA, we must note that the dataset has not yet been collected and compiled. Without actual video recordings and corresponding heart rate data, we cannot fully access its potential inconsistencies, missing values, or biases.

For data consistency, we plan to store the video recordings of the 30 subjects in MP4 file format. After preprocessing, these videos will be converted into frames, which will be stored as JPG files. This approach ensures the dataset maintains consistent formatting.

We also recognize potential biases that may arise from the camera's sensitivity to different skin tones or the impact of external lighting conditions. To address these issues, we will implement algorithmic corrections and conduct test runs to validate the consistency of our data collection.

While the limitations of this project revolve around the dependency on video quality and the subject's stability during the recording process. Any movement may affect the accuracy of the heart rate estimation.

Results and Findings

Since the dataset has not yet been collected and compiled, our focus on data analysis and findings is currently limited to the initial concept and design. We do not have a specific dataset to analyse at this moment. However, once data collection has commenced and the dataset is compiled, we will continuously refine our methods based on initial findings.

In conclusion, the dataset analysis, results and finding for this project are in their formative stages. Our approach includes planning for various steps in data collection, ensuring successful data capture, implementing effective data preprocessing routines, and addressing potential biases and limitations. This strategy sets the groundwork for data analysis once the dataset becomes available.

Tools and Software specifications

	Tools/ Libraries	Descriptions
Programming Language	Python	Optimal for its large libraries essential for data operations, image processing, and machine learning tasks critical for our project.
	Java	Chosen for its integration with Android Studio, facilitating the development of our app within the Android ecosystem.
Model Libraries	NumPy	Essential for numerical computing, enabling efficient handling of arrays and matrices vital for image data processing.
	SciPy	Provides the necessary computational tools for performing analyses like ICA and CCA, essential for our data processing needs.
	OpenCV	A comprehensive library for computer vision tasks, crucial for facial detection and Region of Interest (ROI) extraction.
	scikit-learn	Offers machine learning algorithms that are pivotal for developing predictive models linking facial recognition to heart rate data.
	Matplotlib (with Seaborn)	Enables visual data representation, aiding in analysis and presentation.
	Pandas	Ideal for its data manipulation capabilities, assisting in the preprocessing of complex datasets.
	TensorFlow/PyTorch	Deep learning frameworks that provide advanced capabilities for building accurate neural network models, instrumental for precise heart rate measurement.

Operating system (OS)	Android platform (Mobile)	A prevalent OS for mobile devices, enabling access to a broad user base and diverse hardware integration.
Develop Environment (IDE)	Visual Studio Code	Selected for its adaptability and support for a wide array of development needs and programming languages.
	Android Studio	The definitive IDE for Android app development, offering specialised tools and native platform integration.
Data storage	MySQL	A scalable and reliable RDBMS, ensuring secure and efficient storage of user data and heart rate metrics.
Version control system	GIT	A distributed version control system for tracking changes to software codebase. This enables collaboration among our team members and managing different versions of the algorithm.
Communication	Discord	A communication platform that provides voice, video, and text communication channels.
	WhatsApp	A messaging app that is widely used for quick, informal communication.
	Gmail	Google's email service is for formal communication, document sharing, and easily integrates with other Google Workspace tools.
	Zoom	A video conferencing tool that is ideal for hosting virtual meetings.
Collaboration	Google Drive	A cloud-based storage service that allows for saving files online and sharing them with others.
	Google Docs	A web-based word processor that allows users to create and edit text documents directly in a web browser while collaborating with other team

		members.
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Justification of Choices

A. Programming language

We chose Python as our primary language for algorithm development mainly because of its extensive libraries tailored for scientific computing and data analysis. It has become the go-to choice for a wide range of machine learning tasks, from data analysis to deep learning (Karl, 2023). Essential libraries such as NumPy, SciPy, and scikit-learn provide built-in methods for PCA, CCA, and ICA, which are the key techniques for our project.

Python's simple and straightforward syntax makes it ideal for developing and testing complex algorithms. Moreover, it allows for easy integration with other applications benefiting from the "Write Once, Run Anywhere (WORA)" principle (Advantages and Disadvantages of Python, n.d.).

Besides, we are also using Java for Android development due to its compatibility with the Android Runtime (ART). The choice is further supported by the Android SDK, which offers Java APIs that facilitate direct access to hardware features, such as the device's camera.

B. Operating System

Our decision to focus on the Android platform for the mobile application stems from Android's global market share and its compatibility with a variety of devices, which potentially expands our user base. In 2024, Android dominated the global smartphone market with a 70.69% share, while iOS had a 28.58% market share (Backlinko Team, 2024).

Unlike iOS, which requires actual devices for testing and thus causing a higher cost, Android applications can be tested on Android Virtual Devices at no additional cost. Android's open ecosystem and comprehensive development tools, such as Android Studio, further support our choice.

C. Develop Environment

Our project will make use of the Visual Studio Code for model development and Android Studio for Android mobile app development.

Visual Studio Code supports multiple programming languages, including Python, and offers built-in GIT support to facilitate version control. This integration ensures that our work remains synchronised across the development team.

Android Studio, the official IDE for Android development, includes a powerful emulator for testing applications and tools for optimising app performance through CPU and memory usage analysis. It also supports a range of Android libraries and SDK versions.

D. Database system

We chose to use MySQL because it is open source and free, which helps our project team avoid additional software costs – making it an excellent choice for our project. Moreover, MySQL is compatible across various platforms. In this project, our primary need is to store user historical heart rate data, and our application does not require complex transactions. Since our model is small-scale, MySQL perfectly meets our needs.

An additional benefit is the large community support available online. Whenever we encounter bugs or need solutions, there is always an abundance of resources and community expertise to help us out.

E. Version control

The most well-known version control system is Git, and it will be the version control system that we will use for our project.

Both Visual Studio Code and Android Studio provide built-in Git support, allowing developers to manage version control directly within the IDE. Git offers several features, including a detailed history of changes made by each team member. By using simple push and pull actions, team members can share their work with each other.

The branching and merging capabilities of Git are useful for managing the development of complex algorithms, facilitating experimentation in algorithm development.

F. Communication

We have chosen WhatsApp for routine communications, while Discord and Zoom are reserved for important meetings and discussions. Additionally, Gmail is used for file sharing and important conversations that need to be conducted in a secure manner.

G. Collaboration

Google Drive is our primary tool for storing important documents, accessible from anywhere. Google Docs will serve as our collaborative platform for writing and editing reports, allowing multiple team members to work simultaneously.

Tools and Hardware Specifications

Computer

Hardware	Specifications	Justification
CPU	Minimum requirements: 4 cores processor such as Intel Core i5 or AMD Ryzen 5 Recommended: 8 cores processor such as Intel Core i7 or AMD Ryzen 7	Multi-core processor to handle computational tasks that involve coding, compiling and running simulations or training models for the algorithms. Most machine learning software only requires four threads to run efficiently. Therefore, eight or more cores is recommended for more demanding tasks in deep learning (Delvadiya, 2023).
GPU	Minimum requirements: NVIDIA Geforce GTX1050 Recommended: NVIDIA GeForce GTX1660 Ti	GPUs are specialised technology that can process large amounts of dataset quickly. As a result, they are ideal for graphics rendering, video processing and speeding up complex computations in applications related to AI and machine learning (Anderson 2024). According to Puget System n.d, almost any NVIDIA graphics card will work but newer and

		higher-ends models will offer better performance.
RAM	Minimum requirements: 8GB of RAM Recommendations: 16GB of RAM	At least 8gb of RAM to support multitasking and handle the memory requirements of running multiple development tools such as IDE and web browsers. Small datasets and simple model, such as linear regression or basic decision trees will requires 8GB to 16GB of RAM, medium datasets and moderate model such as random forests or shallow neural networks requires 16GB to 32GB of RAM and large datasets and deep learning is advisable to have 32GB or more RAM (Zaidi, 2023).
Storage	Minimum requirements: 128GB SSD Recommendations: 256GB SSD	SSD uses NAND flash memory therefore speed is one of its main advantages. SSD provide far quicker read and write rates since they store and retrieve data using flash memory, unlike the traditional hard drive disk (Geonix, 2023)

Camera

Hardware	Specifications	Justification
Canon EOS 1500D Kit (EF S18-55 IS II)	Frame rate: 1920 x 1080 (full HD) 30p and 1280 x 720 (HD) 60p	Higher frame rate per second(FPS) is the measurement of the amount of pictures being taken per second by the camera. Therefore, higher FPS can provide higher accuracy and performance (Cyberlink, 2021). Higher frame rate provides more temporal information and improves tracking accuracy for image processing tasks.
	Resolution: (720 x 480) pixel to (6000 x 4000) pixel	Higher image resolution, the more pixels are bunched together that create a smoother and more detailed image (Adobe, n.d). This is important after getting ROI and performing calculations for the heart rate measurement.

Heart Rate Measurement Tools

Hardware	Specifications	Justification
Zacurate 500C Elite Fingertip Pulse Oximeter	Price: \$34.99 Accuracy: +/-2 bpm Measure SpO2 and Pulse Rate (PRbpm)	The wearable pulse oximeter can measure heart rate and oxygen saturation in the red blood cell. It can achieve medical level uses whereas Apple itself highlighted that the sensors used in the watch are not for medical grade (Viatom, 2023). Therefore, we chose oximeters because it is more cost-effective and of its simplicity and ease of use.

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